PROJECT REPORT

ECOMMERCE CUSTOMER BEHAVIOR ANALYSIS

College: Govt. College for Women, Karnal

Batch: Batch-1 (25090)

Project ID: SIC/BD/016

Project Link : https://github.com/radhakalra18/BIG_DATA

Dataset: USA Ecommerce Data (Jan 3, 2014 – Dec 30, 2017)

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E-Commerce Customer Behavior Analysis using Spark

Abstract

The analysis of the sample dataset provides valuable insights into business performance across different dimensions such as products, customers, and regions. It highlights key factors influencing sales and profit, including discounts, product categories, and geographical trends.

The implementation of data processing using PySpark ensures scalability and efficiency in handling large datasets. Visualization techniques help in better understanding patterns and trends, making the analysis more interpretable for decision-makers.

Furthermore, predictive modeling techniques demonstrate the potential of machine learning in forecasting future sales and profit. Among the models used, ensemble methods such as Random Forest and Gradient Boosting show improved performance compared to basic regression models.

Overall, this project successfully demonstrates how data analytics and machine learning can be leveraged to support strategic business decisions. It lays a strong foundation for building more advanced, real-time, and intelligent analytics systems in the future.

Introduction

E-commerce platforms generate large volumes of transactional data. Big data technologies such as Apache Spark and PySpark allow companies to process

and analyze this data efficiently. Customer behavior analysis helps organizations understand purchasing trends, customer segments, and factors

influencing sales performance. This report analyzes a retail dataset to identify patterns and insights.

Objectives of the Study

1. To analyze customer purchasing patterns.
2. To identify the most profitable product categories.
3. Build predictive models for forecasting.
4. To analyze regional sales performance.
5. To observe monthly sales trends.

Dataset Description

The dataset used in this project is the “sample” retail dataset which contains transactional data from an e-commerce platform. Each row in the dataset presents a unique order transaction containing customer information, product details, geographic location, and financial metrics such as sales, discount, and profit. The dataset is widely used for business analytics, customer behavior analysis, and data visualization projects. It contains thousands of transaction records across multiple product categories and regions. Understanding the structure and datatype of each attribute is important for performing correct analysis using big data frameworks such as PySpark.

Column Name	**Data Type**	**Description**
Row ID	int64	Unique identifier for each row
Order ID	object	Unique order identifier
Order Date	object	Date when the order was placed
Ship Date	object	Date when the order was shipped
Ship Mode	object	Shipping mode used for delivery
Customer ID	object	Customer identifier
Customer Name	object	Customer name
Segment	object	Customer segment type
Country	object	Country of the customer
City	object	City of the customer
State	object	State of the customer
Postal Code	int64	Postal code of delivery area
Region	object	Sales region
Product ID	object	Unique product identifier
Category	object	Product category
Sub-Category	object	Product category

Product Name	object	Product name
Sales	float64	Total sales value of transaction
Quantity	int64	Quantity of items ordered
Discount	float64	Discount applied on product
Profit	float64	Profit earned from transaction

Methodology (PySpark Approach)

Typical PySpark workflow used in big data environments:

1. Data Loading using Spark DataFrame
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis(EDA)
4. Model Building
5. Model Evaluation and Visualization

Example PySpark Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("EcommerceAnalysis").getOrCreate()
df = spark.read.csv("sample.csv", header=True, inferSchema=True)
df.printSchema()
df.describe().show()
```

Tools and Technologies Used

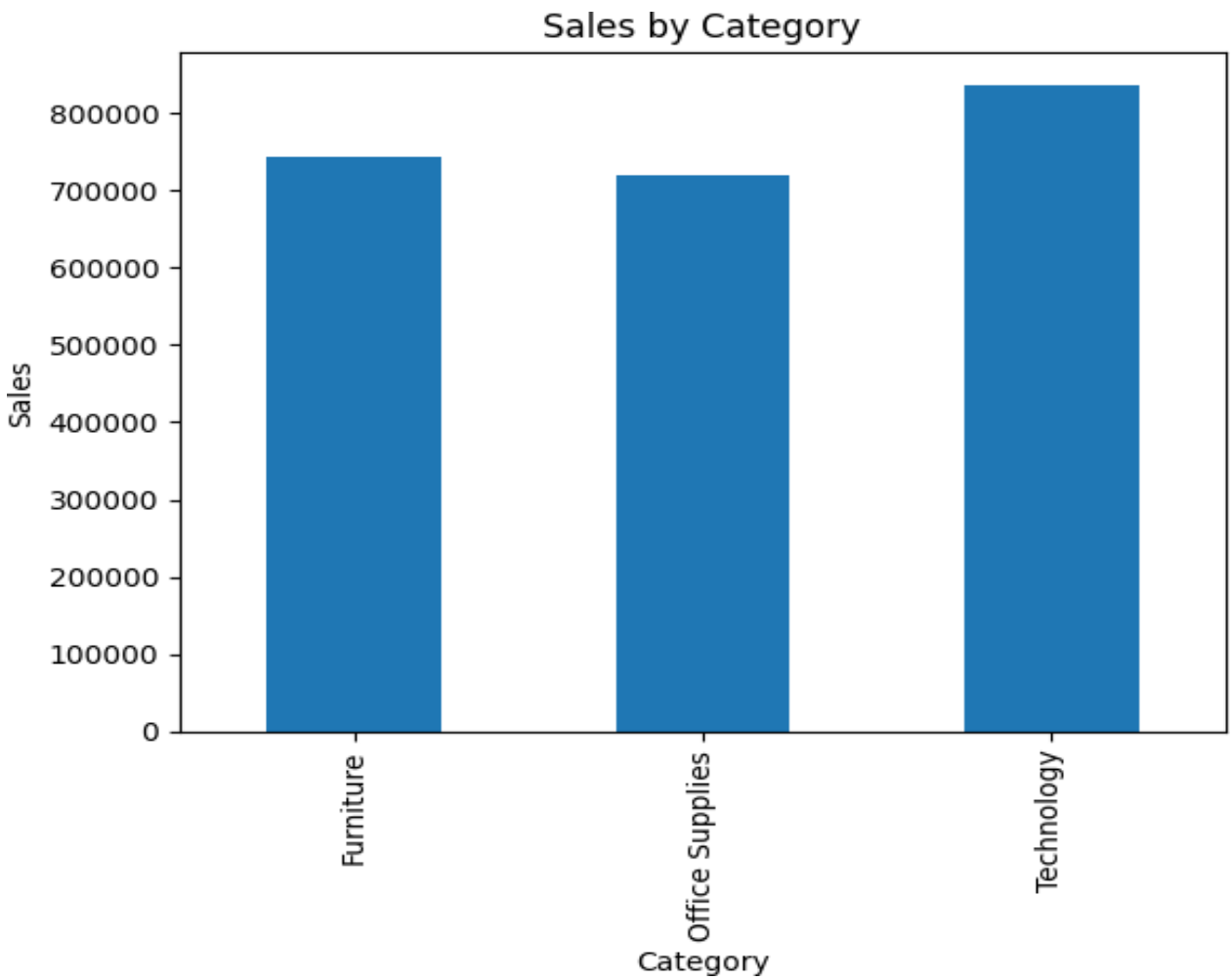
- Kaggle
- PySpark
- Pandas
- Plotly

- Machine Learning Algorithms
- Google Colab

Sales by Category Analysis

Product category analysis helps businesses understand which product types generate the highest revenue.

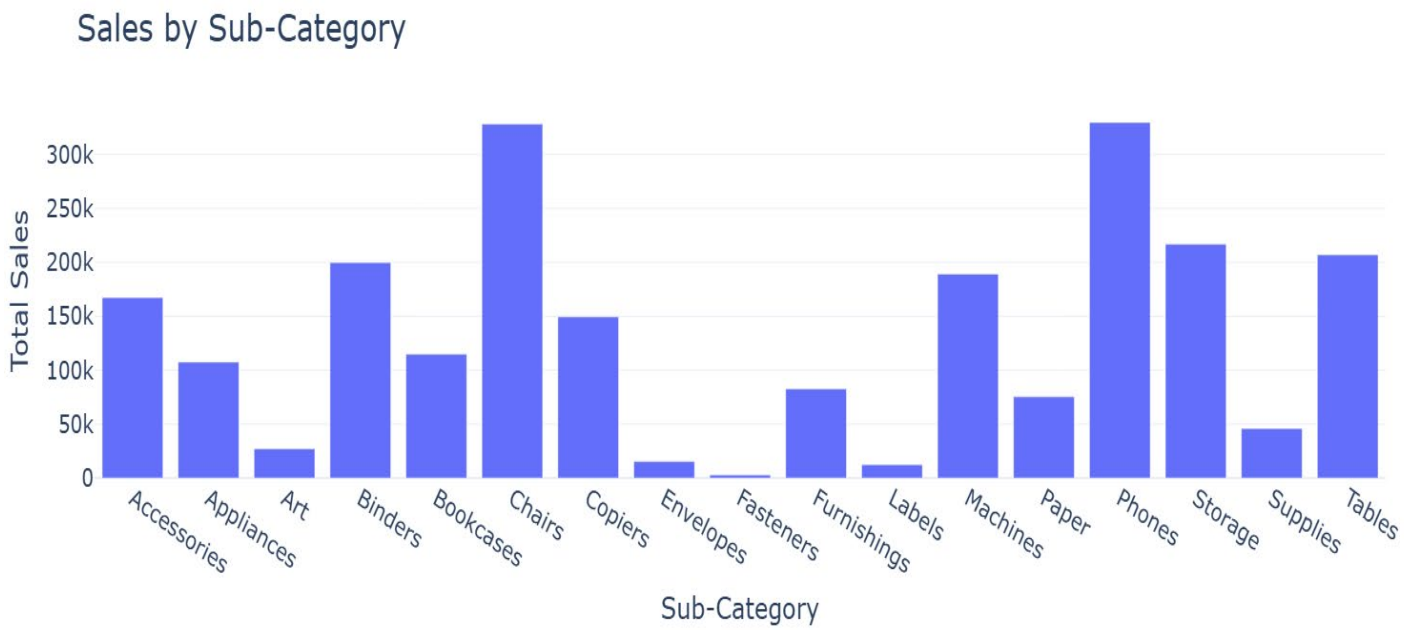
The visualization below illustrates total sales by product category.



Sales by Sub-Category

The chart shows that **Chairs and Phones** generate the highest sales, followed by **Storage, Tables, and Bookcases**. In contrast, categories like **Fasteners and Envelopes** have the lowest sales.

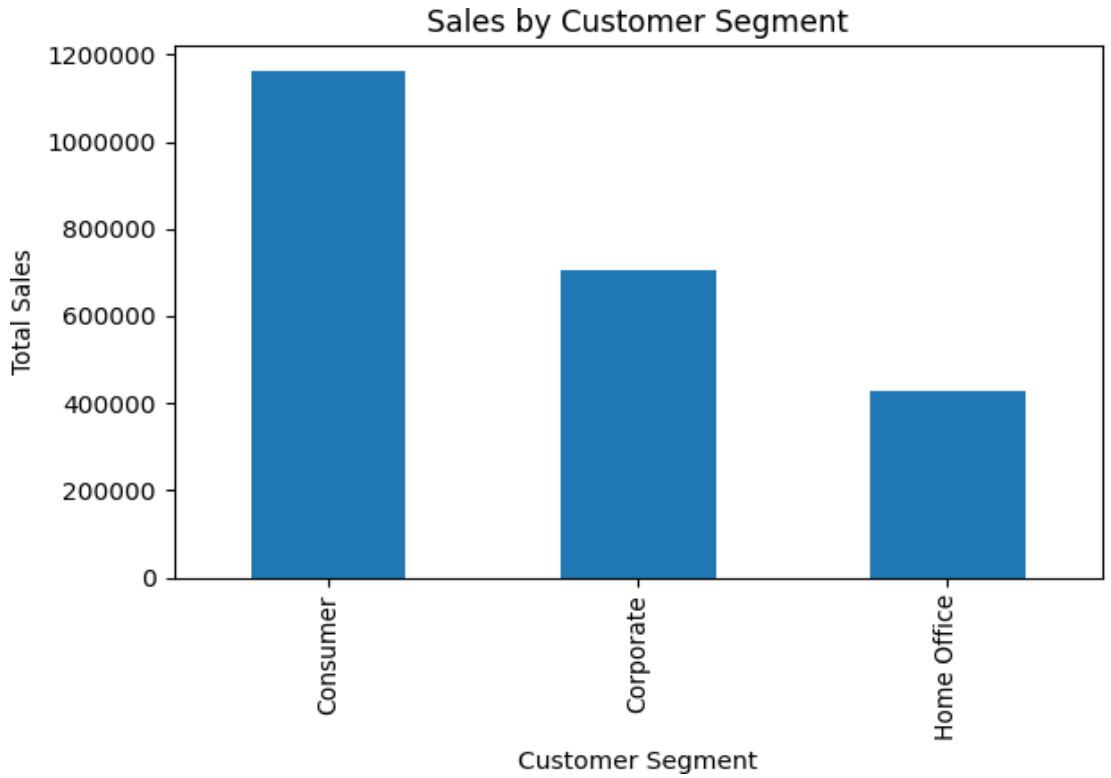
Insight: Sales are concentrated in a few high-performing categories, indicating that furniture and technology products drive most of the revenue.



Customer Segment Analysis (Sales based)

Customer segmentation divides customers into groups based on purchasing behavior. The dataset contains three major segments: • Consumer • Corporate • Home Office

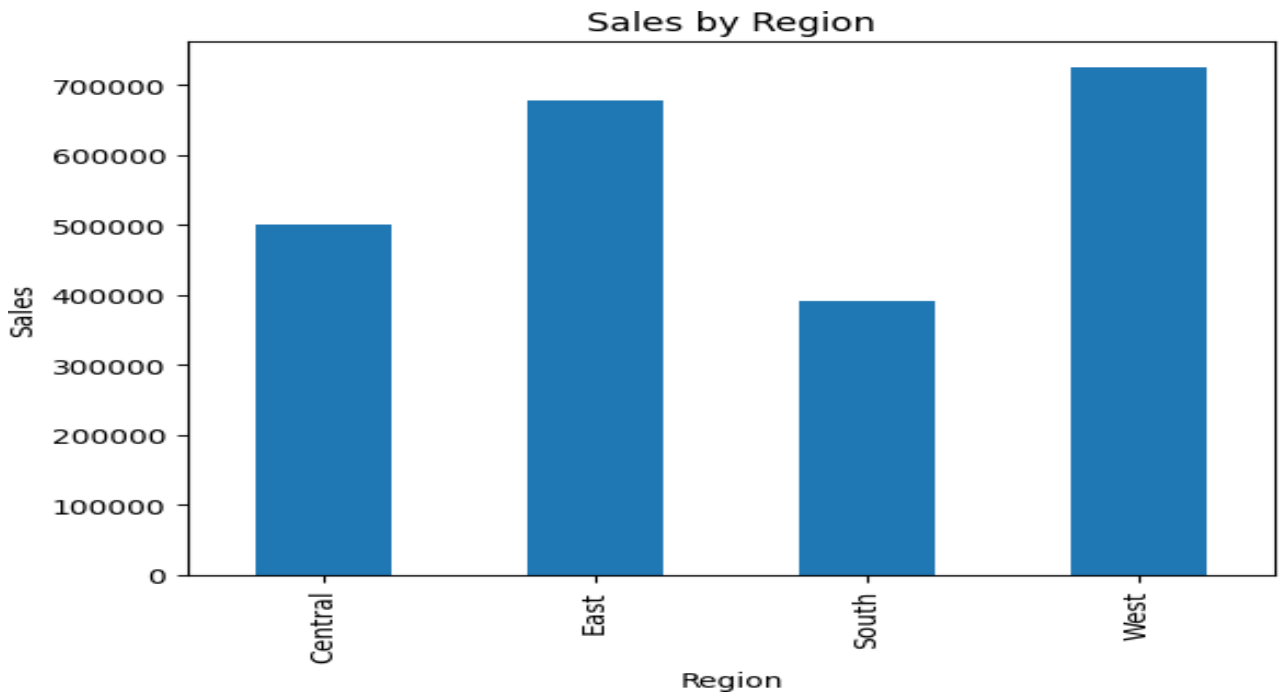
Businesses can tailor marketing strategies based on these segments to improve sales and customer satisfaction.



Regional Sales Analysis

Regional sales analysis identifies geographic areas contributing most to total revenue.

Companies can use these insights to optimize logistics and regional marketing strategies.



Monthly Sales Trend

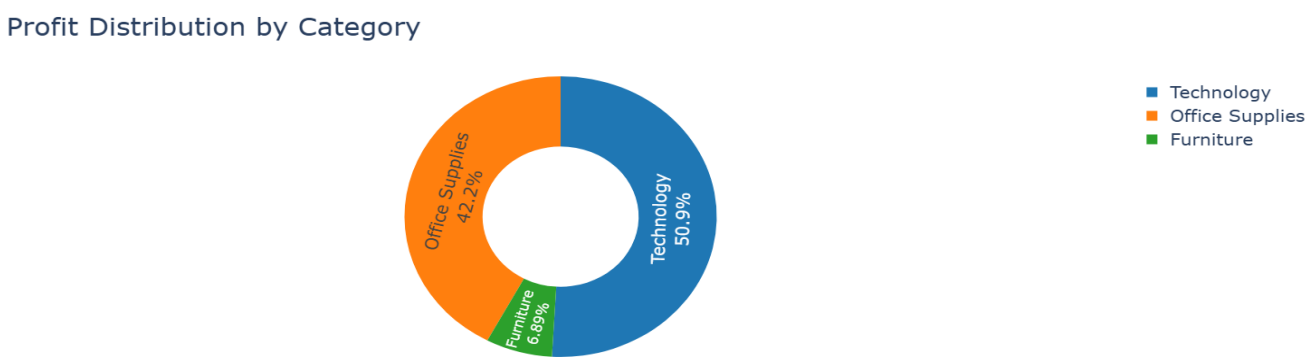
Sales trend analysis helps businesses understand seasonal patterns and fluctuations in demand.

This allows companies to prepare inventory and marketing campaigns accordingly.

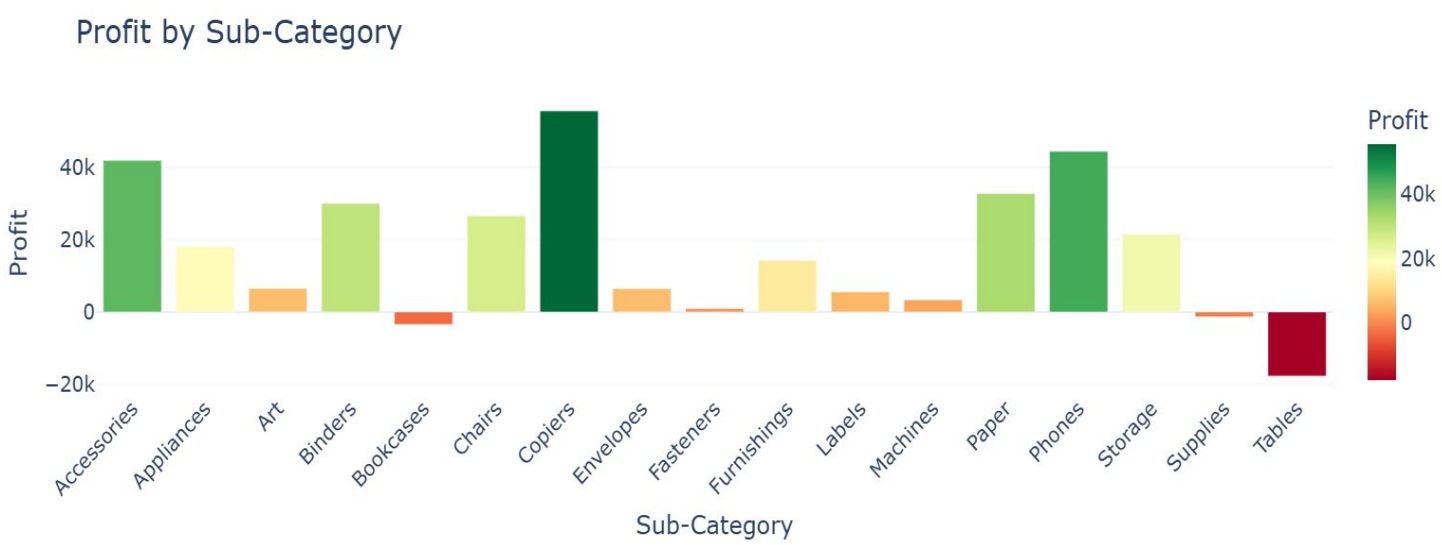
Profit Analysis

- **Monthly Profit Trend :** Profit shows an overall upward trend, with fluctuations in the middle of the year and peak performance in the final months, especially December.

- **Profit by Category :** Electronics generates the highest profit, followed by Clothing, while other categories contribute less.



- **Profit by Sub-Category :** Accessories, Phones, and Copiers are the top-performing sub-categories, whereas Tables, Chairs, and Machines show low or negative profit.

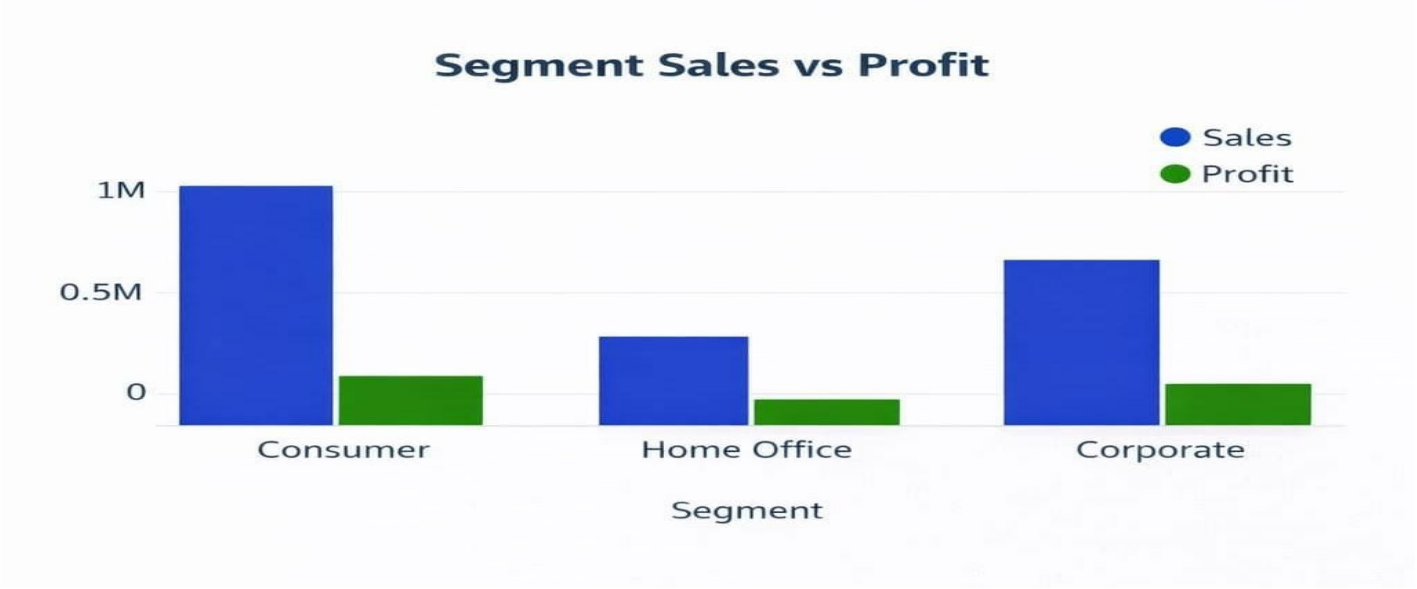


Customer Segment Sales vs Profit

The chart compares sales and profit across different customer segments. The Consumer segment generates the highest sales and profit, making it the most valuable segment for the business. The Corporate segment follows with moderate performance, while the Home Office segment records the lowest sales and profit.

Overall Insight

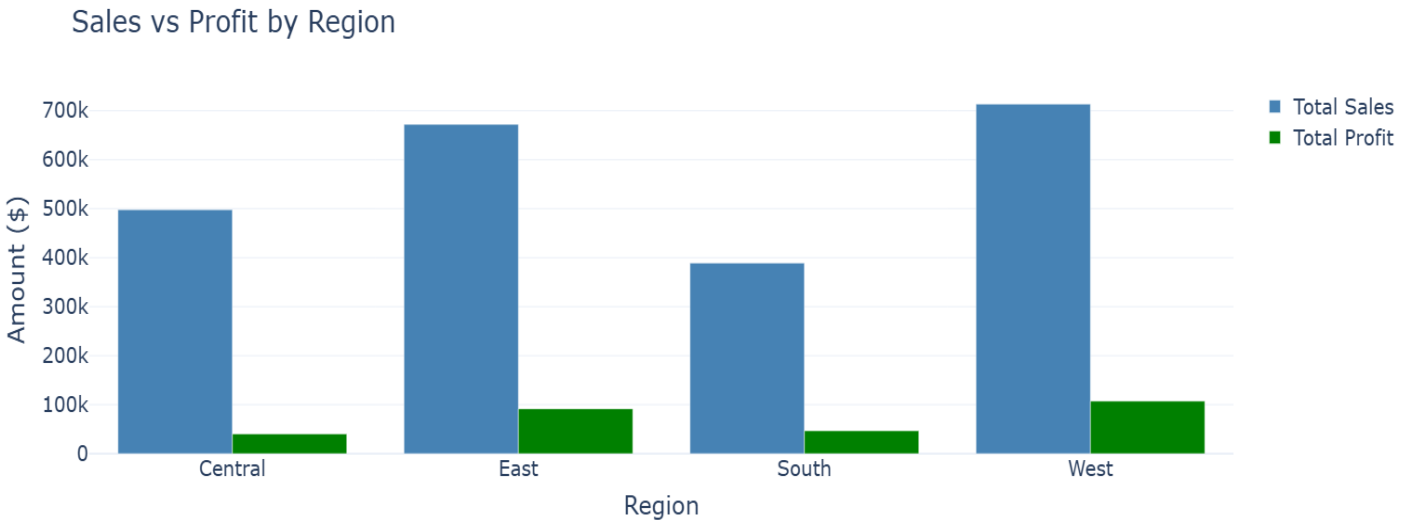
The business is highly dependent on the Consumer segment, indicating a need to strengthen strategies for Corporate and Home Office segments to achieve balanced growth.



Region Sales vs Profit

The chart illustrates sales and profit performance across different regions. The West and East regions generate the highest sales and profits, indicating strong market performance. The Central region shows moderate sales with relatively lower profit, while the South region records the lowest sales and profit among all regions.

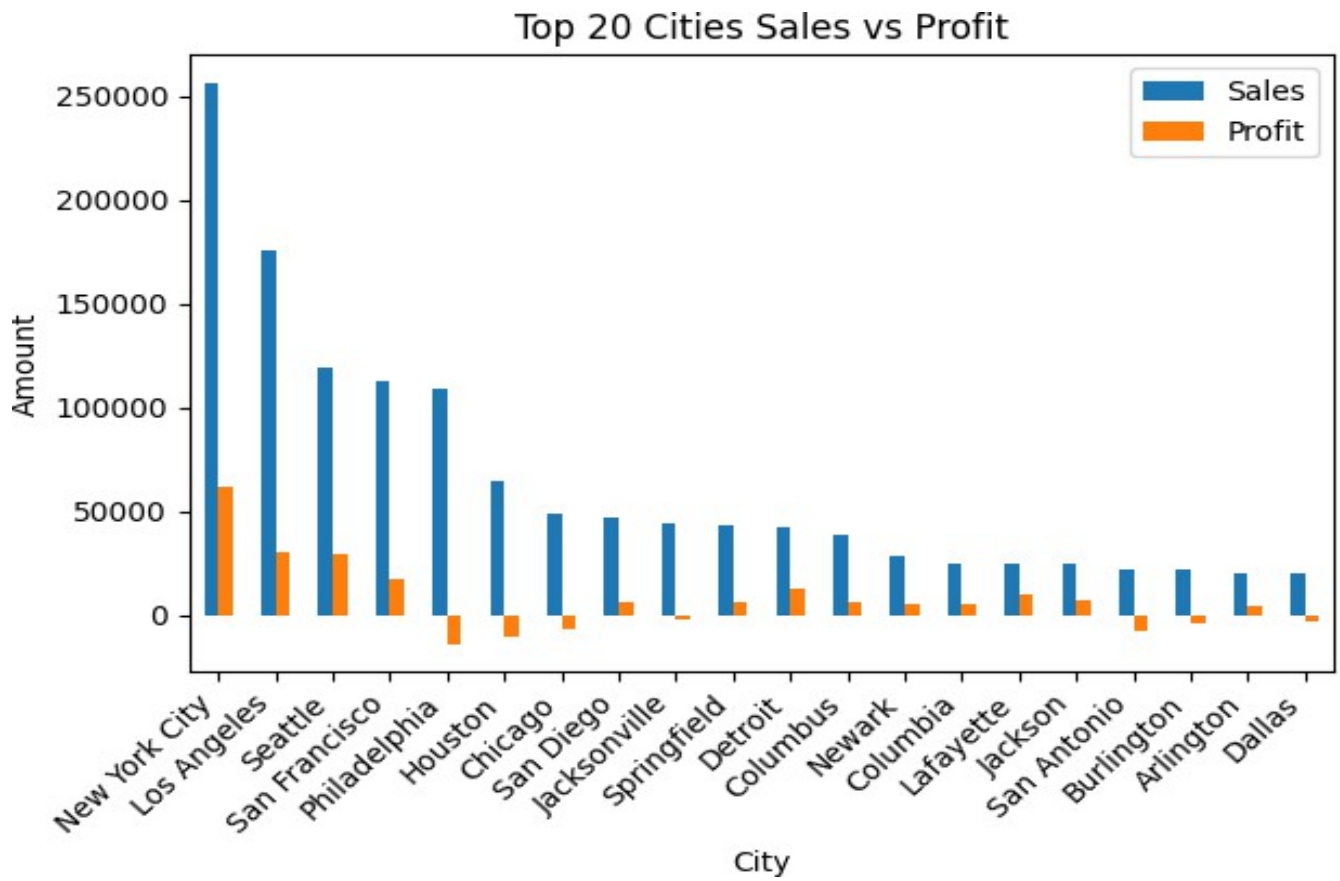
- **Overall Insight**
  The business performs best in the West and East regions, while the South region requires improvement strategies to enhance both sales and profitability.



Top 20 Cities Sales vs Profit

The chart compares sales and profit across the top 20 cities. New York City and Los Angeles lead in both sales and profit, making them the most profitable markets. Cities like Seattle, San Francisco, and Philadelphia also show strong performance with relatively high sales and positive profit.

The business performs best in major metropolitan cities, while certain cities require better cost management to improve profitability.



Top Product Affinity Rules

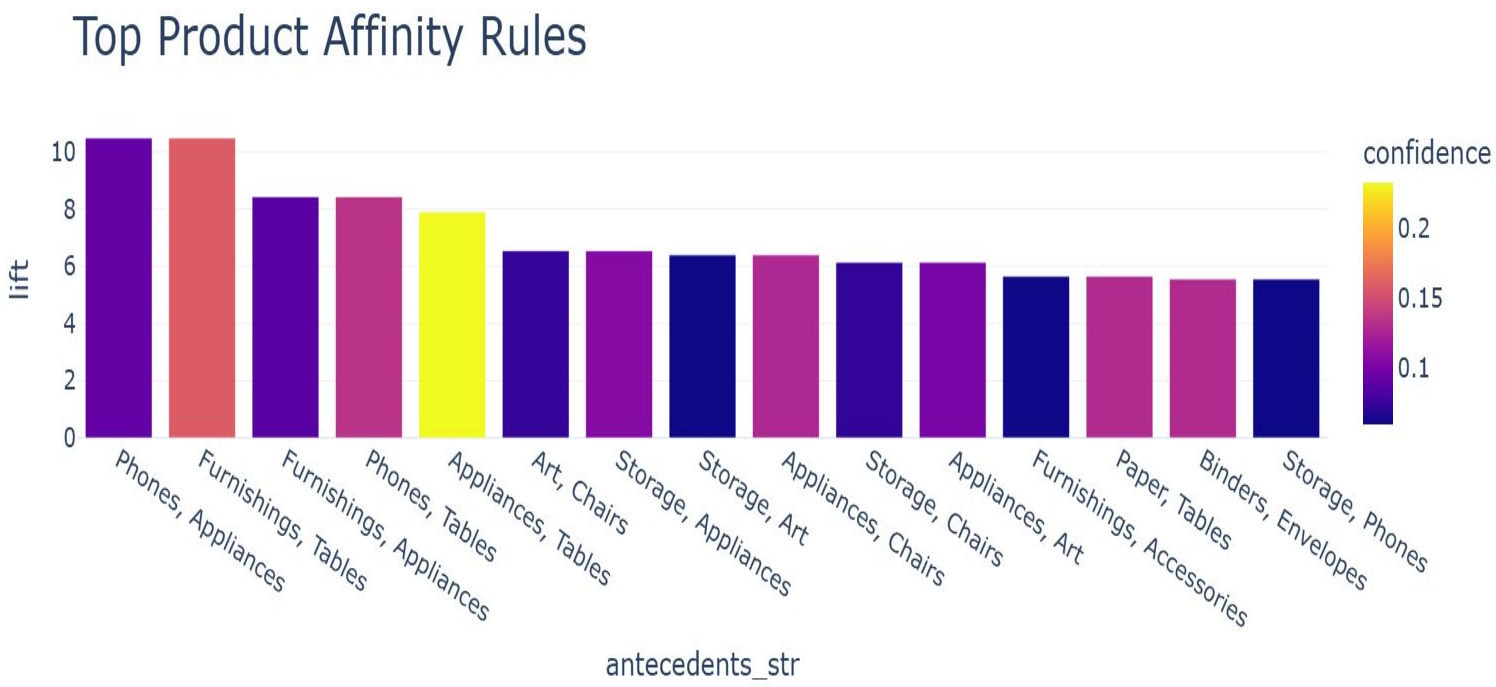
The chart shows strong product combinations based on lift values, indicating items frequently bought together. Pairs like **Tables → Furnishings** and **Appliances → Phones** have the highest association.

Confidence Insight :

Higher color intensity represents stronger confidence, meaning more reliable purchasing patterns.

Overall Insight :

These insights help in improving cross-selling strategies and product bundling to increase sales.

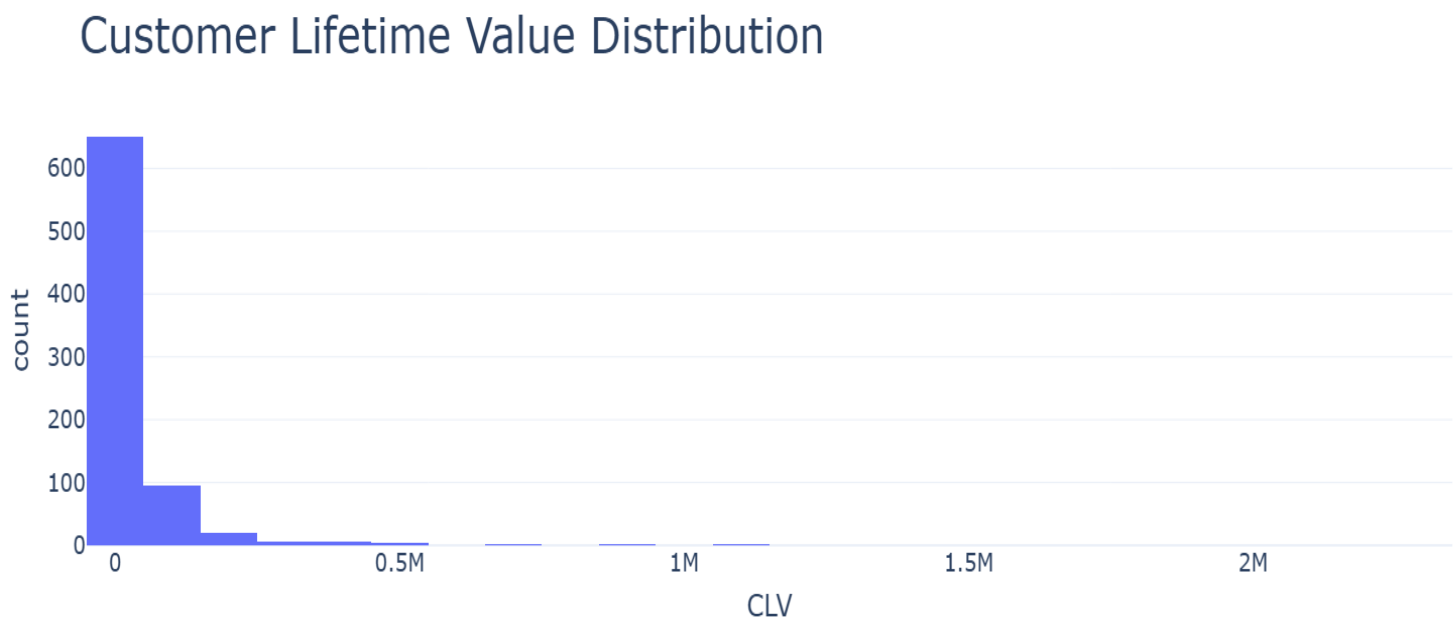


Customer Lifetime Value (CLV) Distribution

The CLV distribution chart shows a highly right-skewed pattern, where the majority of customers have a low lifetime value, while a small group contributes significantly higher value.

- Most customers fall within the lower CLV range, indicating limited long-term spending.
- A few customers exhibit extremely high CLV, acting as key revenue drivers.
- The long tail suggests the presence of high-value loyal customers, though they are relatively rare.

Insight:
Businesses should focus on retaining high-value customers and implementing strategies to increase the CLV of low-value segments through targeted marketing and engagement.



Model Prediction Comparison

The chart compares the performance of different machine learning models—Linear Regression, Random Forest, and Gradient Boosting (GBT)—against the actual customer spend.

Model Performance:
All models closely follow the downward trend of actual spend, indicating their ability to capture overall patterns. Gradient Boosting (GBT) shows the closest alignment with actual values, especially in the initial and middle ranges. Linear Regression performs consistently but with slight deviations, while Random Forest shows more fluctuations and variability.

Conclusion

The analysis of the sample dataset provides valuable insights into business performance across different dimensions such as products, customers, and regions. It highlights key factors influencing sales and profit, including discounts, product categories, and geographical trends.

The implementation of data processing using PySpark ensures scalability and efficiency in handling large datasets. Visualization techniques help in better understanding patterns and trends, making the analysis more interpretable for decision-makers.

Furthermore, predictive modeling techniques demonstrate the potential of machine learning in forecasting future sales and profit. Among the models used, ensemble methods such as Random Forest and Gradient Boosting show improved performance compared to basic regression models.

Overall, this project successfully demonstrates how data analytics and machine learning can be leveraged to support strategic business decisions. It lays a strong foundation for building more advanced, real-time, and intelligent analytics systems in the future.

Future Scope

The current analysis provides valuable insights into sales, profit trends, and customer behavior using historical transactional data. However, there are several opportunities to extend and enhance this project in the future:

1. Advanced Predictive Modeling

Future work can include implementing more advanced machine learning and deep learning models such as XGBoost, LSTM, or Neural Networks to improve prediction accuracy for sales and profit forecasting.

2. Customer Segmentation & Personalization

Applying clustering techniques (e.g., K-Means, DBSCAN) can help segment customers based on purchasing behavior, enabling personalized marketing strategies and targeted promotions.

3. Real-Time Data Processing

Integrating real-time data pipelines using tools like Apache Kafka or Spark Streaming would allow businesses to monitor sales and customer activity dynamically and make quicker decisions.

4. Geospatial Analysis

Enhancing the analysis with geospatial visualization (maps) can provide deeper insights into region-wise performance and help optimize logistics, delivery, and regional marketing strategies.

5. Inventory & Supply Chain Optimization

Future scope includes integrating inventory data to predict demand, reduce stockouts, and optimize supply chain operations using predictive analytics.

6. Interactive Dashboard Development

Developing a fully interactive dashboard using tools like Power BI, Tableau, or advanced Plotly Dash applications can improve usability for business stakeholders.

7. Recommendation System

Building a product recommendation system using collaborative filtering or content-based filtering techniques can enhance customer experience and increase sales.

8. Cloud Deployment

Deploying the solution on cloud platforms (AWS, Azure, or Google Cloud) can improve scalability, accessibility, and performance for large-scale business applications.

9. A/B Testing & Business Experiments

Future work can include running controlled experiments to evaluate pricing strategies, discount effectiveness, and marketing campaigns.

10. Data Security & Governance

Implementing data governance, privacy policies, and secure data pipelines will be essential when scaling the system for real-world applications.